AI CLUSTERS

Multi-planar AI Scale Out Fabrics

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Introduction to Multi-Planar Fabrics

Theory

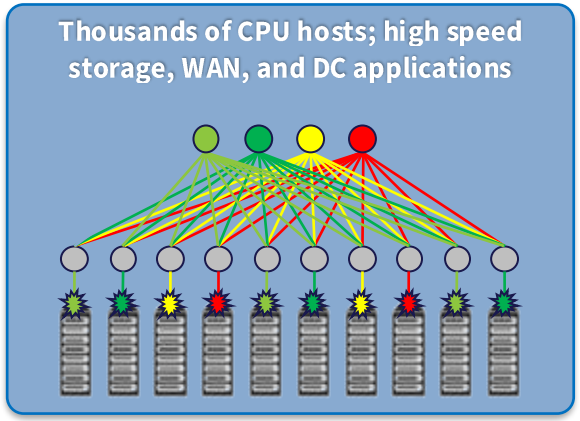
- What is a multi-planar fabric?
- Why would you want one?
- Variants
- Implementation challenges

Practice

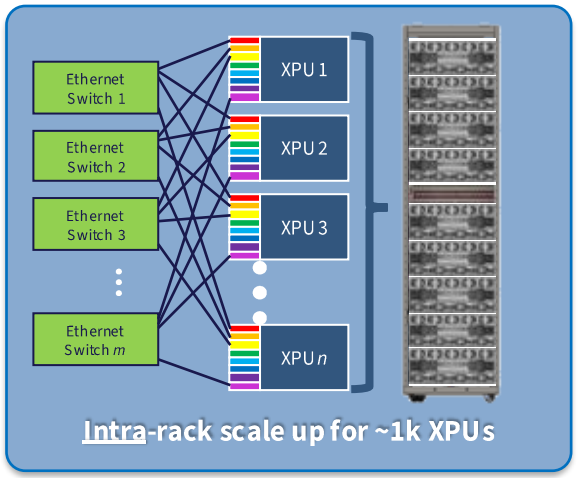
- Rack-scale AI with AMD Helios and Open Rack Wide (ORW)
- Multi-planar fabrics with AMD Pensando Vulcano and Multipath Reliable Connection (MRC)
- Building a large-scale cluster with Arista 7060XE7

AI Fabrics

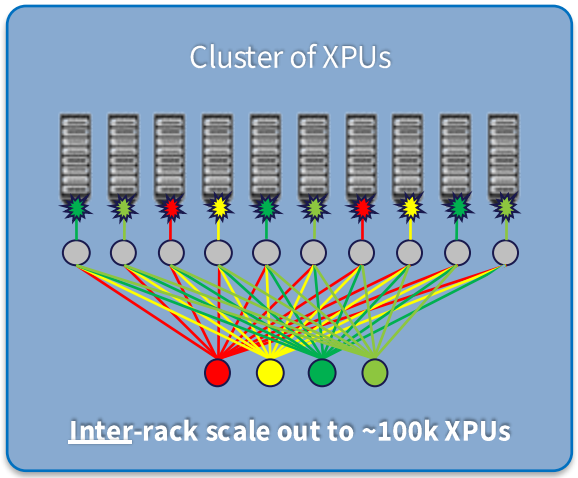
Front End AI Data Center



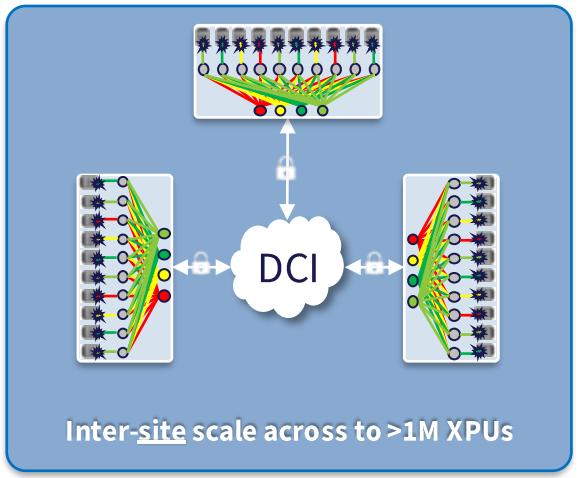
The 3 AI Fabrics



Scale Up



Scale Out

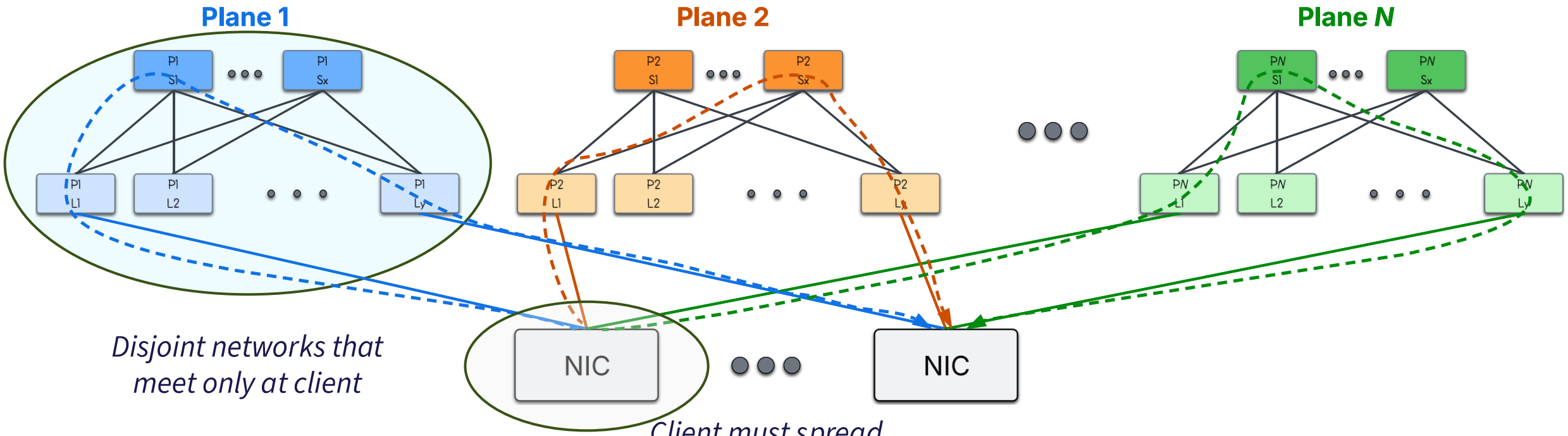


Scale Across

AMD

ARISTA

Multi-planar Fabrics



Benefits

Redundancy

- Planes provide redundancy and strong isolation
- However, relies on *client* to manage plane health because networks are disjoint

Scale and Cost

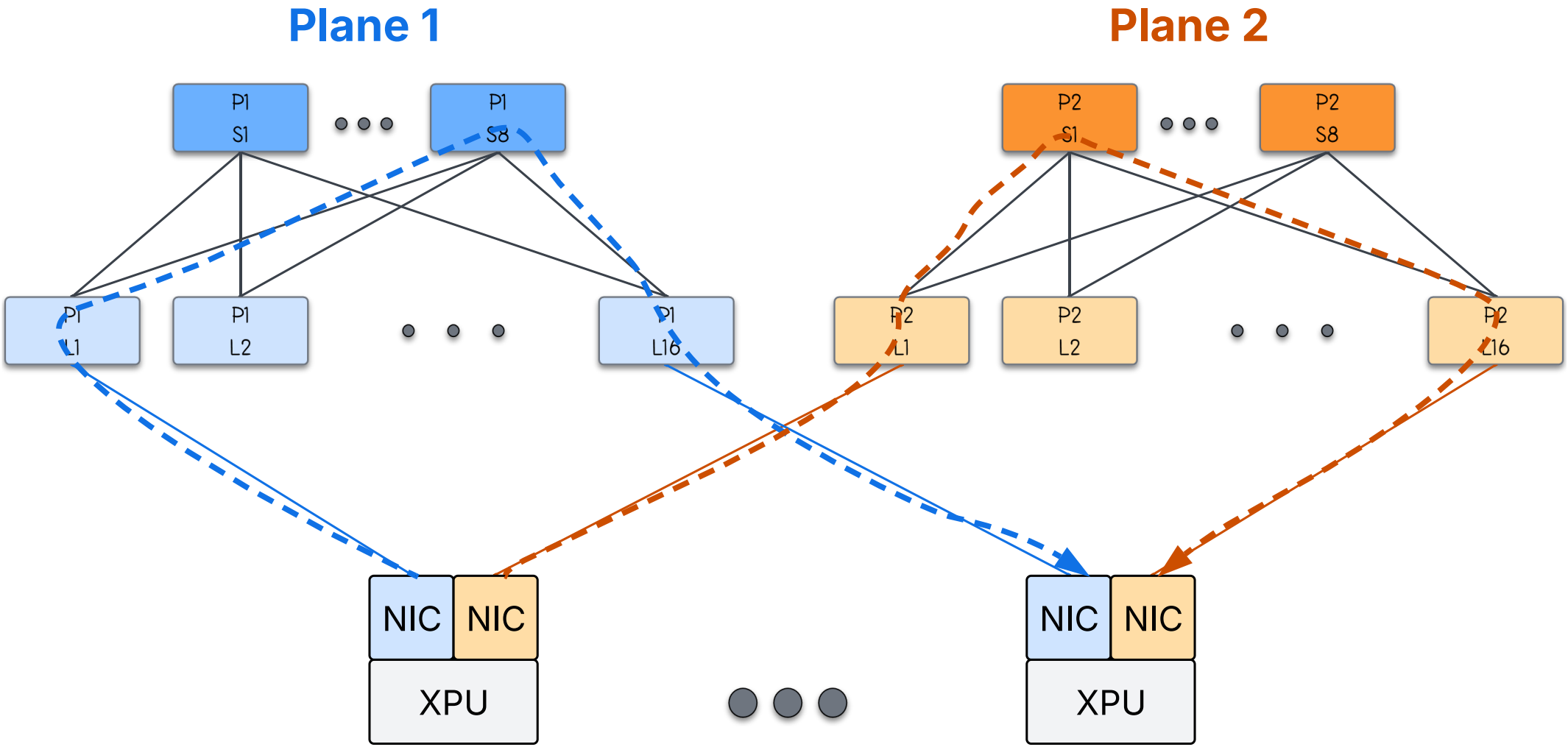
- Network scale depends on device radix (k), which increases as port speed decreases

=> Single-tier networks scale further: 100s or 1,000s of XPUs before you need two tiers

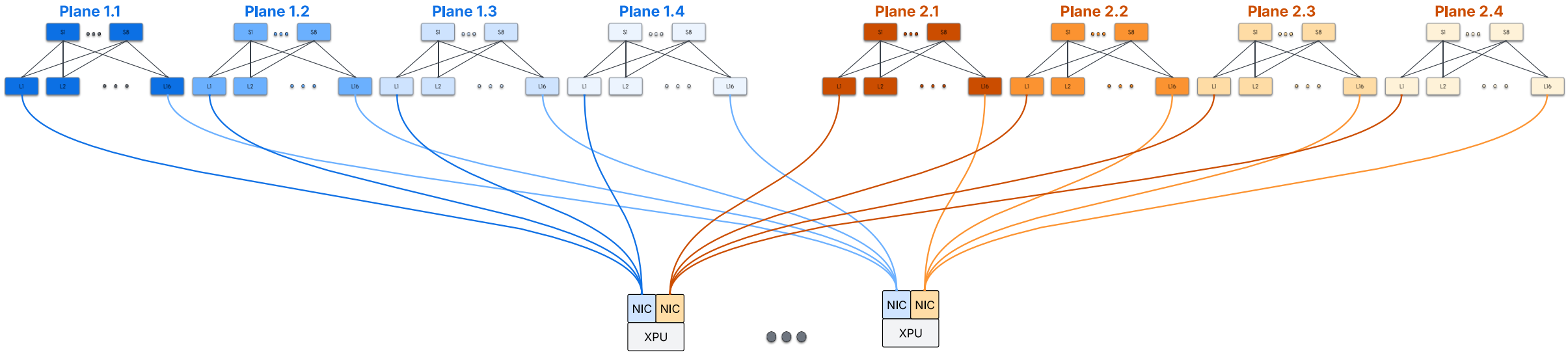
=> Two-tier networks scale *much* larger, so no need for three tiers

  - Two-tier Clos network (1:1) provides up to $k^2/2$ client ports, so benefit scales with *square* of radix
  - With 102.4T devices:
    - @800G: device radix is 128 -> maximum scale is 8,192 clients
    - @200G: device radix is 512 -> maximum scale is 131,072 clients

Variant: Multi-planar with multiple NICs



Combination: Multi-planar at Scale



Implementation challenges (1)

Load Balancing and Fault Isolation

- Client must *distribute load* across disjoint planes
- Client must *handle out-of-order* packet delivery
- Client must *detect failures* and redistribute load
  - Local cases are easier (e.g., link to first hop is down)
  - Remote cases are more difficult (e.g., one specific destination is unreachable via a single plane)
  - Partial failures can be very challenging – congestion or loss in the fabric

Solutions

- For simple cases (e.g., dual NIC) => system software stack may implement
- Otherwise, typically implemented in NIC hardware and firmware
  - Open solutions are available:
    - OCP *Multipath Reliable Connection* (MRC) – for AI workloads
    - *Ultra Ethernet Transport* (UET) – for a broader set of AI, HPC, and RDMA workloads
  - Both MRC and UET go beyond minimum need and implement fully adaptive multipath solutions

Implementation challenges (2)

Optics and Cabling

- Physical layer must support required plane fanout
  - Single NIC (or cage) may require connectivity to 1, 2, 4, or even 8 separate leaf switches

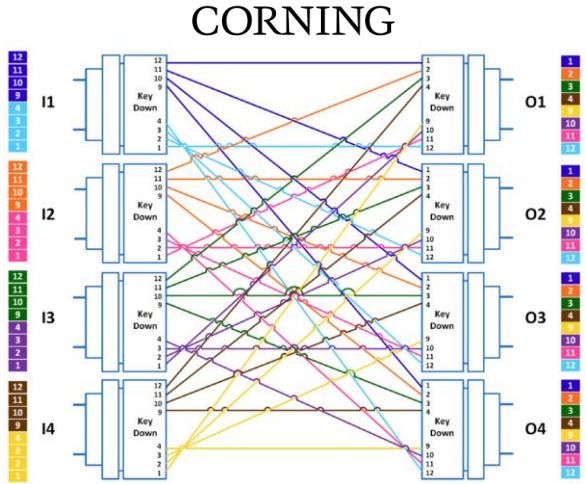
For up to two planes per NIC: Dual-port optics

- Many common 800G and 1.6T optics are available with two physical connectors
  - Common types: dual-MPO (2VSR, 2DR4) and dual-LC (2FR4)
  - For many scenarios, this provides ample scale and enables simple cabling



For greater than two planes per NIC: Parallel optics and fiber shuffles

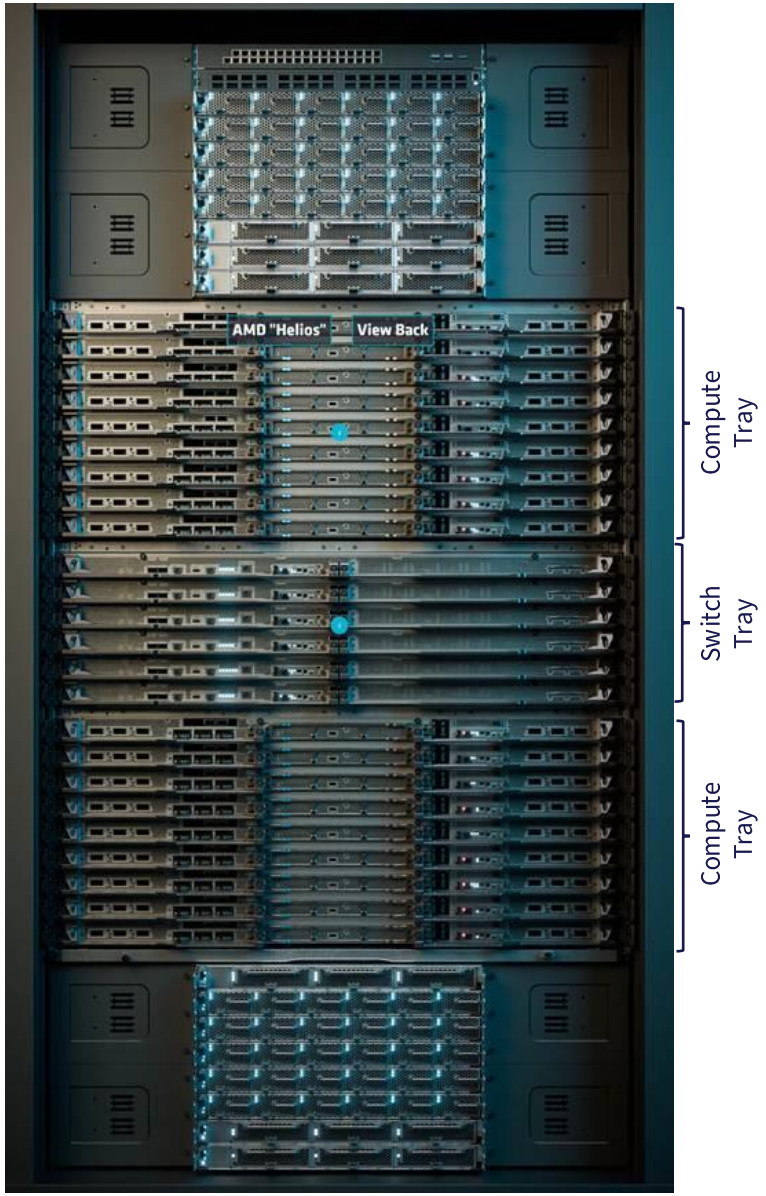
- Larger scales require >2 planes per transceiver
- Requires optical PMDs based on parallel fibers (DR or SR)
- Fibers broken out to achieve required fan out and then shuffled to fan in
- Cabling can be complex
  - Smaller deployments can use shuffle cables
  - Larger deployments typically build fiber shuffling into the physical infrastructure



Rack-scale AI

AMD Helios – OCP ORW based Rack Scale GPU System

- Rack Components
  - 18x 1 OU
    - 1P:4GPUs compute trays
  - 6x 1 OU switch trays
  - 44OU x 1200mm wide x 1289mm deep
- Scale up network
  - 1.8 TB/s BW per GPU
  - UALoE using industry stand ethernet switching
  - Single switch hop, multi-plane topology
- Front-End network options
  - Salina DPU as FE NIC
  - Vulcano FE-NIC

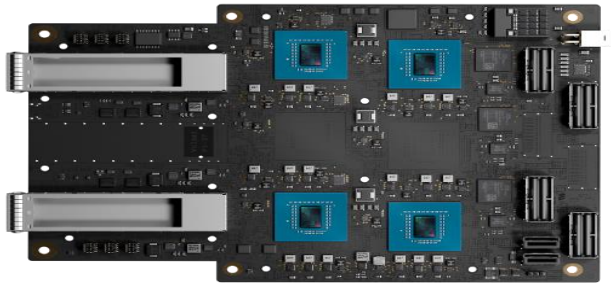


Scale Out Network: AI NIC

- One or Two 800G Vulcano NIC(s) per GPU
- Given 1.6T OSFP cage is shared across two GPUs
- RHS optics
  - 1x800G (100G or 200G SerDes)
  - 1x1.6T-DR8 or 1.6T-2xDR4 OSFP (200G SerDes)
- Plane Topologies
  - Plane Per NIC – Single or Dual Plane Topologies
  - Plane Per SerDes – Up to 8 Plane Topologies
- Network Load Balancing – Enough Entropy in packet
  - NIC based packet spraying
  - **NIC based source routing – SRv6-uSID**
  - NIC based in-order placement – Network Packet Spray



OCP 3.0 TSFF
1x800G NIC per GPU



Custom FF
2x800G NIC per GPU

MRC (Multipath Reliable Connection)

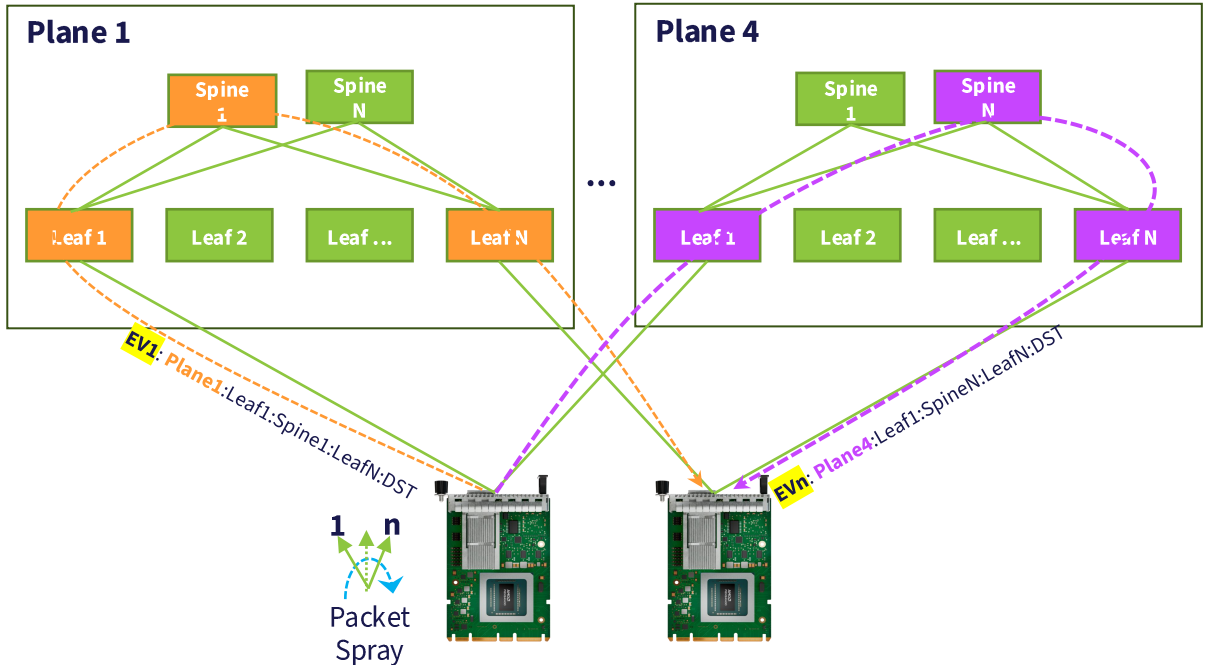
Entropy Value (EV)-based packet spraying
- Both Controller based and Generated EV support
- Each QP's EV profile is split across planes — every EV maps to one fixed plane
- Per-EV state (GOOD / SKIP) — used or avoided based on observed path congestion

Source NIC inserts SRv6 uSID stack for end-to-end source routing
- Traffic follows a specific plane and a specific path within that plane
- No ECMP or dynamic load balancing (DLB) needed in the fabric — path is fully specified at the source

Congestion signaling from the fabric
- Switches mark ECN or trim packets to signal congestion to NICs
- Receiver NIC relays these signals back to the source NIC via SACK/NACK

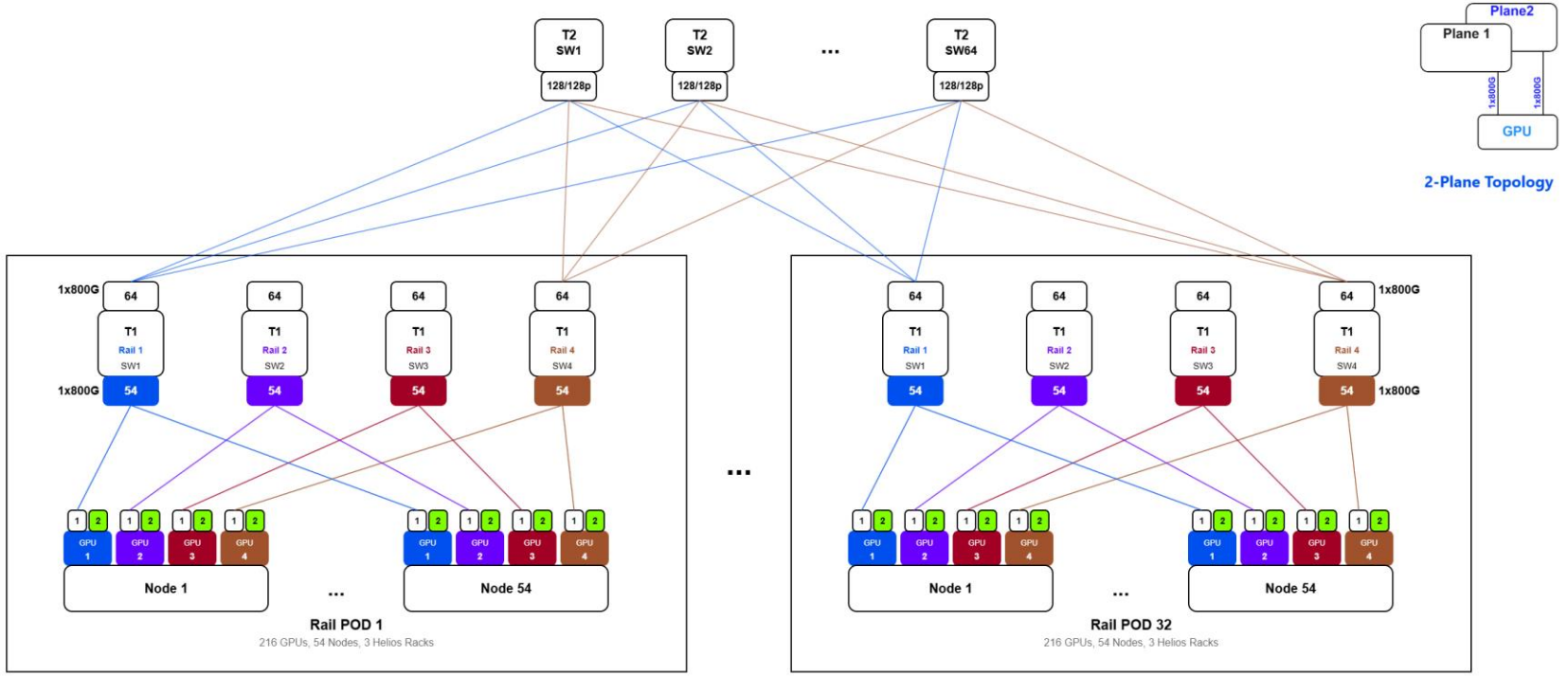
Congestion control
- driven by ECN + RTT monitoring

Out-of-order delivery
- packets placed directly to memory as they arrive, no reordering required



Plane Per NIC Topology

- 6,912 GPUs, 96 Helios Racks
- 2x800G per GPU
  - 2 Planes
    - Each serving 800G
- 4 Rails
- Modular Rail Pod or SU
  - 216 GPUs per Rail POD
  - 3xHelios Racks

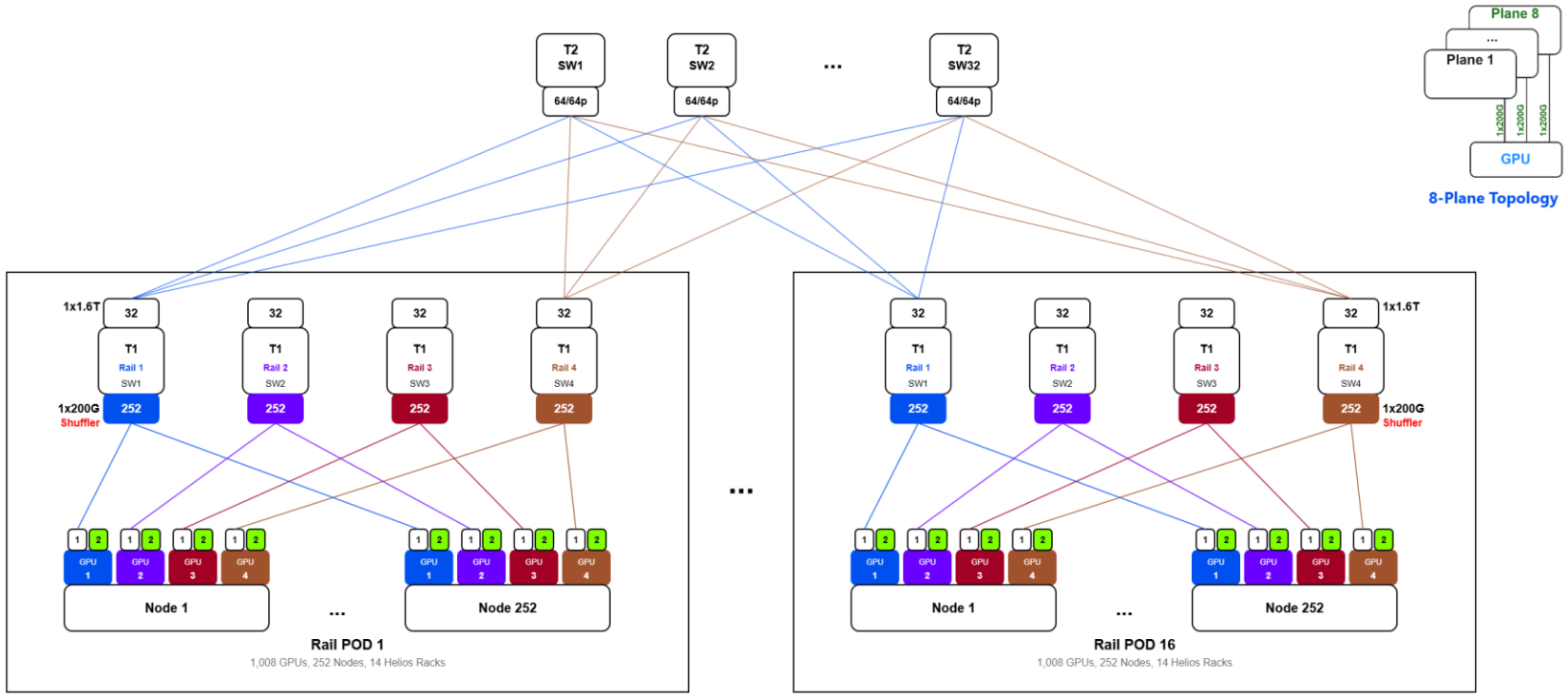


6,912 GPUs | 1,728 Nodes | 32 Rail PODs × 3 Helios Racks | 2 Planes

TH6 (102.4T) Liquid | DL: 128p (800G) | UL: 128p (800G) | T2: 64 Fixed (128/128p, 1x800G/T1) | Ratio: 0.84:1

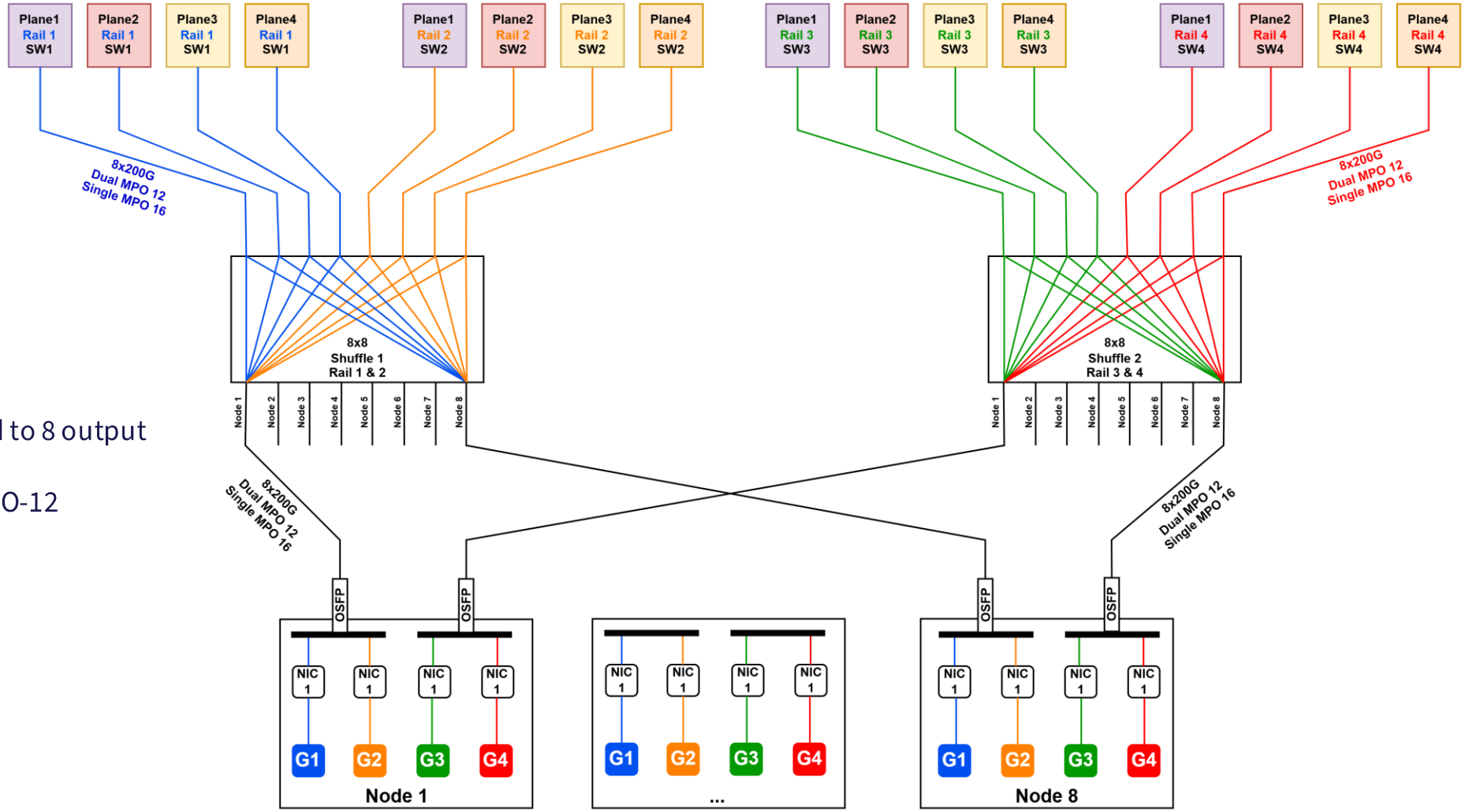
Plane Per SerDes Topology

- 16,128 GPUs, 224 Helios Racks
- 2x800G per GPU
  - 8 Planes
    - Each serving 200G
- 4 Rails
- Modular Rail Pod or SU
  - 1,008 GPUs per Rail POD
  - 14xHelios Racks
- Max GPU scale: 129,024 in 2 Tier
- Requires shuffler between GPU and T1s



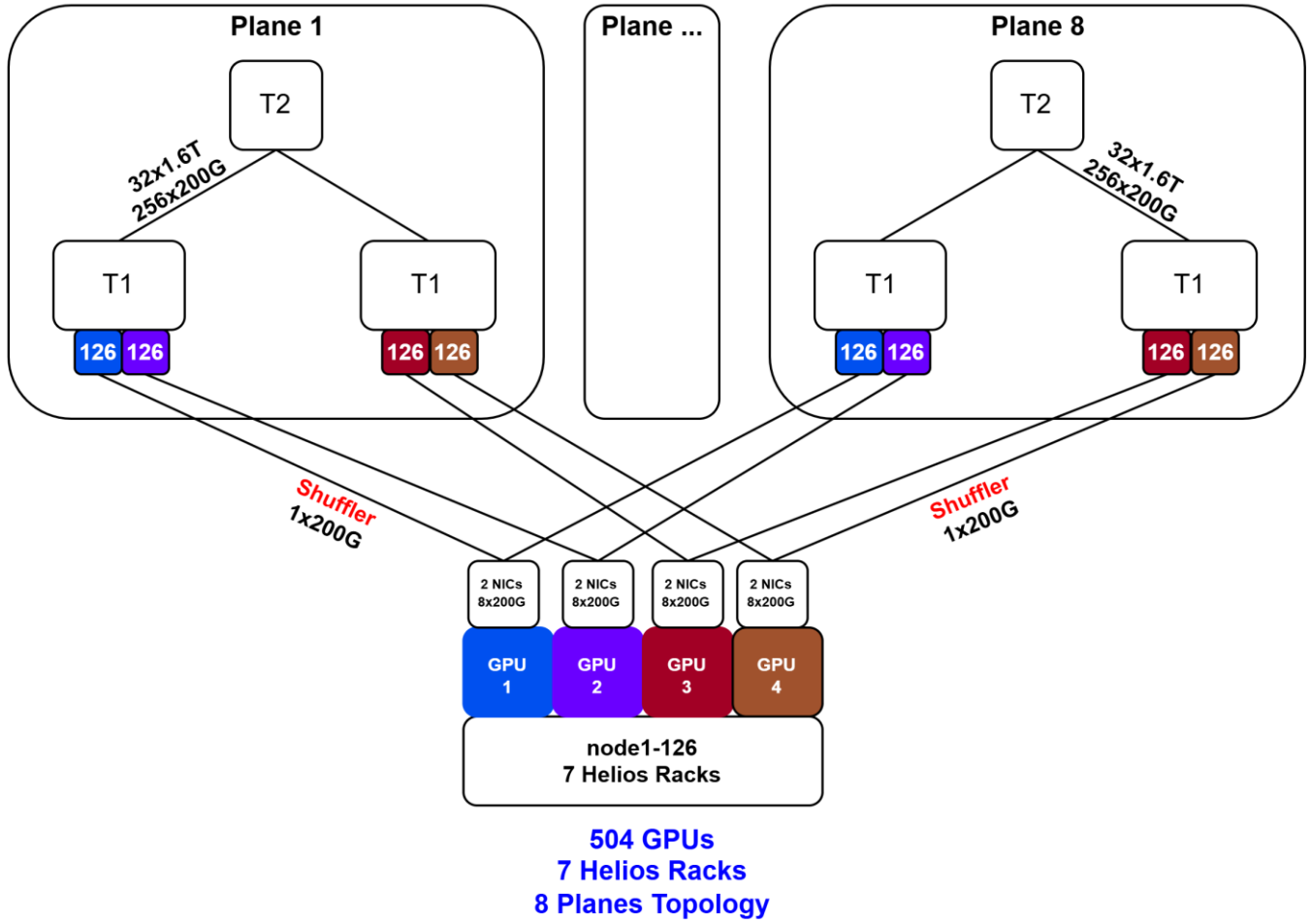
Shuffler

- 8x8 shuffler
  - 8 inputs x 8 outputs
  - Each input port 8x200G shuffled to 8 output ports
- GPU NIC to shuffle MPO-16 or Dual MPO-12



Validation Cluster

- 504 GPUs, 7 Helios Racks
- Tomahawk6-200G
  - Arista 7060XE7-EOS
- Plane Per SerDes Topology
  - 8 Planes - each 200G
  - 1.6T-DR8 Optics
  - 8x8 shuffler between T1 and GPUs
- 4 Rails
- MRC
  - Native IPv6
  - SRv6-uSID
  - 128 or 256-ECMP



Call to Action

- Evaluate multi-planar AI Fabrics for your next build
  - Benefits include redundancy, cost, and scale
- Explore next-generation multi-path, multi-planar open transport protocols for AI
  - *OCP Multipath Reliable Connection (MRC)* and/or *Ultra Ethernet Transport (UET)*
  - Support an open ecosystem by requesting support from your suppliers
- Links
  - High-Performance Networking with AMD and Arista:
    - https://www.arista.com/assets/data/pdf/Whitepapers/AMD-Next-Gen-AI-White-Paper.pdf
  - OCP MRC Specification: https://www.opencompute.org/documents/ocp-mrc-1-0-pdf
    - Arista White Paper: https://www.arista.com/assets/data/pdf/Whitepapers/MRC-Whitepaper.pdf
  - Ultra Ethernet Specification: https://ultraethernet.org/wp-content/uploads/sites/20/2026/07/UE-Specification-1.0.3.pdf
    - Arista White Paper: https://www.arista.com/assets/data/pdf/Whitepapers/Demystifying-Ultra-Ethernet-WP.pdf

Thank You!

